

MACHINE LEARNING-BASED LATE DELIVERY RISK PREDICTION IN GLOBAL SUPPLY CHAIN OPERATIONS

Business Analytics and Predictive Supply Chain Research Paper

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Project: Machine Learning Based Late Delivery Risk Prediction in Global Supply Chain Operation

Internship Project Report

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EXECUTIVE SUMMARY

This project, titled “Machine Learning–based Late Delivery Risk Prediction in Global Supply Chain Operations,” focuses on identifying potential shipment delays before they occur and supporting proactive logistics decision-making. Late deliveries can lead to service-level agreement violations, increased operational costs, customer dissatisfaction, and inefficient use of logistics resources. Therefore, early identification of high-risk orders is important for improving delivery performance.

The project uses a supply chain transaction dataset containing 180,519 records. The target variable and `Late_delivery_risk`, indicates whether an order was delivered late or on time. The analysis combines data preparation, exploratory data analysis, SQL-based analysis, predictive modelling, and Tableau-based business intelligence visualization.

The analysis identified 98,977 late-delivery records, representing an overall late-delivery rate of approximately 54.83%. Shipping mode analysis showed substantial differences in delivery performance. First Class recorded the highest late-delivery rate at 95.32%, followed by Second Class at 76.63%, Same Day at 45.74%, and Standard Class at 38.07%. These results indicate that delivery risk varies significantly across different shipping methods.

The project also examines delivery risk across regions, markets, customer segments, and region–shipping mode combinations. Predictive techniques such as Logistic Regression and Random Forest are used within the proposed modelling framework to estimate the probability of late delivery at the order level. Each order can then be classified into Low, Medium, or High Risk categories to support operational prioritization.

Interactive Tableau dashboards are used to present the major findings through delay-risk KPIs, shipping-mode comparisons, regional risk analysis, and high-risk order lists. These dashboards are designed to help operations teams quickly identify areas requiring attention and support faster decision-making.

The findings suggest that logistics organizations can benefit from adopting a risk-based approach to shipment management. High-risk orders can be prioritized for early review, while high-risk regions and shipping modes can be monitored more closely. Proactive customer communication, improved shipment planning, and better allocation of operational resources can further reduce the impact of potential delivery delays.

Overall, the project demonstrates how supply chain analytics and predictive modelling can support a transition from reactive delay management to proactive delivery-risk intelligence. The integration of SQL analysis, machine learning, and Tableau visualization provides a practical framework for improving shipment visibility, operational planning, and data-driven decision-making in global supply chain operations.

ABSTRACT

Late delivery is a persistent operational risk in global supply chains because missed service commitments can create SLA breaches, penalty costs, customer dissatisfaction, and avoidable last-minute intervention. Traditional reporting explains delays only after they occur; operational teams require an earlier mechanism that identifies risky shipments before fulfillment decisions are finalized. This study develops a machine-learning-oriented late-delivery risk framework using the public DataCo SMART Supply Chain dataset and presents the analysis in the business context of global logistics operations.

The analytical workflow combines SQL-based data validation and exploratory analysis, leakage-aware feature selection, supervised classification, order-level risk scoring, and Tableau visualization. The target variable is `Late_delivery_risk`, where 1 indicates a late delivery and 0 indicates an on-time delivery. A cleaned analytical/scored dataset containing 180,519 order-item records was used in the project workflow. Predictive variables were restricted to information that can reasonably be available before the realized delivery outcome, while outcome-revealing variables such as actual shipping duration and delivery-status fields were treated cautiously to avoid data leakage.

Descriptive analysis shows that 98,977 of 180,519 records are marked as late, corresponding to an overall late-delivery rate of approximately 54.83%. Shipping mode is a major operational differentiator: First Class records a 95.32% late rate, Second Class 76.63%, Same Day 45.74%, and Standard Class 38.07%. The final operational design assigns each record a Late Delivery Probability and a Low, Medium, or High Risk category, supporting prioritization through an interactive dashboard and a high-risk order action panel.

The study demonstrates how descriptive analytics and predictive risk intelligence can be combined to move supply-chain management from reactive delay handling toward proactive intervention. The proposed workflow supports risk-based prioritization, targeted review of shipping-mode and regional combinations, and more timely operational action.

Keywords: Late Delivery Risk, Supply Chain Analytics, Machine Learning, Random Forest, Logistics Analytics, Predictive Analytics, Tableau, Risk Scoring.

1. INTRODUCTION

Global supply chains operate across multiple markets, customer segments, product categories, and transportation choices. This complexity creates uncertainty in delivery performance. When an order misses its promised delivery window, the consequences can extend beyond a single shipment: firms may face service-level agreement violations, additional expediting cost, customer complaints, reduced trust, and a higher probability of customer attrition.

Most conventional logistics dashboards are retrospective. They report how many shipments were delayed, where delays occurred, and how historical performance changed. Although such reporting is useful, it does not directly answer the forward-looking operational question: which orders are likely to be delayed before they are shipped? Predictive analytics addresses this gap by converting order and logistics information into an estimated probability of late delivery.

This project, titled “Machine Learning-Based Late Delivery Risk Prediction in Global Supply Chain Operations,” was undertaken as a business analytics internship project through Unified Mentor. The case framing uses APL Logistics (KWE Group) as the operational stakeholder context. The empirical analysis itself is based on the public DataCo SMART Supply Chain dataset published by Constante, Silva, and Pereira (2019), ensuring that the research is based on an open and reproducible source rather than proprietary company records.

The project integrates SQL, machine learning, and Tableau. SQL is used to validate the data, check missing values and duplicates, review category consistency, and perform exploratory business analysis. Machine learning is used to convert pre-shipment features into a probability of late delivery. Tableau is then used to transform those predictions into operational KPIs, risk distributions, regional and shipping-mode views, and a prioritized high-risk action table.

The central contribution of the project is therefore not only a prediction model, but an end-to-end decision-support workflow: data preparation → risk prediction → probability scoring → risk classification → dashboard monitoring → operational prioritization.

2. PROBLEM STATEMENT AND OBJECTIVES

2.1 Problem Statement

Global logistics operations often respond to delays after the shipment has already missed its expected service window. This reactive approach increases the cost of corrective action and limits the ability of operations teams to reroute, reprioritize, communicate with customers, or allocate resources before service failure occurs. The business problem is the absence of an early-warning mechanism that quantitatively identifies shipments with elevated late-delivery risk.

The project therefore focuses on predicting late-delivery risk from order-time and shipment-planning information, generating an interpretable risk score for each record, and presenting the results in a dashboard that operations teams can use to prioritize intervention.

2.2 Research Objectives

1. To measure the overall incidence of late delivery in the analytical dataset.
2. To analyze late-delivery patterns across shipping modes and other operational dimensions.
3. To prepare a leakage-aware set of pre-shipment predictor variables for machine learning.
4. To use a supervised classification approach to estimate the probability of late delivery for individual records.
5. To classify orders into Low, Medium, and High Risk categories for operational prioritization.
6. To create Tableau dashboards for overall risk monitoring, region-and-mode analysis, and an operations action panel.
7. To convert analytical findings into practical recommendations for proactive logistics decision-making.

3. DATASET DESCRIPTION

3.1 Dataset Source and Scope

The research uses the DataCo SMART Supply Chain for Big Data Analysis dataset, published in 2019 on Mendeley Data by Fabian Constante, Fernando Silva, and António Pereira. The public dataset contains structured supply-chain records covering provisioning, sales, commercial distribution, customer information, product information, financial measures, order geography, and shipping outcomes. The original dataset contains more variables than are required for predictive modeling; the project therefore creates a cleaned analytical/scored extract for modeling and dashboard use.

The working analytical/scored file contains 180,519 order-item records. The project treats the data at order-item level; therefore, the number of rows should not automatically be interpreted as the number of unique customer orders.

3.2 Target Variable

The target variable is Late_delivery_risk. It is a binary classification label:

- Late_delivery_risk = 1: the record is associated with a late delivery.
- Late_delivery_risk = 0: the record is associated with an on-time or non-late delivery.

Because this target represents the delivery outcome to be predicted, model inputs must be restricted to variables available before the outcome is known.

3.3 Key Variables Used in the Project

Variable	Business Meaning
Days for shipment (scheduled)	Promised/planned shipping duration.
Category Name / Category Id	Product category information.
Customer Segment	Consumer, Corporate or other customer grouping.
Market / Order Region / Order Country	Geographical market and fulfillment location.
Order Item Discount / Discount Rate	Price concession applied to the order item.
Order Item Product Price / Product Price	Product pricing information.
Order Item Quantity	Number of units ordered.

Variable	Business Meaning
Sales / Order Item Total	Commercial value associated with the transaction.
Order Profit Per Order / Profit Ratio	Order-level profitability measures.
Shipping Mode	Service mode such as First Class, Same Day, Second Class, Standard Class.
Late_delivery_risk	Binary prediction target.
Late_Probability	Model-generated probability of late delivery (0–1).
Risk_Category	Operational classification: Low, Medium, or High Risk.

4. RESEARCH METHODOLOGY

4.1 Overall Analytical Framework

The project follows an applied business analytics workflow designed to connect data quality, predictive modeling, and operational decision support. The sequence is:

Data Import → SQL Validation and Cleaning → Exploratory Data Analysis → Leakage Review → Feature Preparation → Model Development → Probability Scoring → Risk Classification → Tableau Dashboard → Operational Recommendations

4.2 SQL-Based Data Validation and Preparation

The dataset was imported into a relational SQL environment and reviewed before modeling. The validation process included row-count checks, schema and data-type inspection, duplicate checks, missing-value checks, inconsistent text review using trimming and case normalization, numerical-range validation, and target-distribution analysis. SQL was also used to calculate descriptive delivery KPIs and shipping-mode late rates.

This stage is important because predictive performance is meaningful only when the underlying analytical table is structurally consistent and the target distribution is correctly understood.

4.3 Data Leakage Control

A critical methodological issue in delivery prediction is data leakage. A feature creates leakage when it contains information that is only known after the prediction moment or directly reveals the target outcome. In a pre-shipment use case, the model should not depend on variables such as Days for shipping (real) or realized Delivery Status because these describe what actually happened after fulfillment. Using them may produce unrealistically high model performance that would not be available in real operations.

Accordingly, the predictive framework emphasizes features that can be known at or before shipment planning, including scheduled shipping days, shipping mode, market/region, customer segment, category, quantity, discount, price, sales, total, and order profitability variables. Recent research on the same DataCo benchmark similarly emphasizes point-in-time feature auditing and leakage-aware evaluation.

4.4 Feature Engineering

The project requirement identified several business-oriented predictive indicators that can be created from the available data.

These include:

- Shipping pressure indicators that relate promised shipping time to order characteristics such as quantity or order complexity.
- Mode-risk flags that distinguish service modes with historically different delay tendencies.
- Regional or market-level congestion indicators based on historical delivery-risk patterns.
- Order-complexity indicators constructed from quantity, category, value, discount, and other order attributes.

Categorical variables can be encoded using one-hot encoding or another suitable method. Numerical features may be scaled where required by the selected algorithm. Class imbalance can be handled through class weighting or resampling when justified by the model-development results.

4.5 Model Development

The project specification includes Logistic Regression as an interpretable baseline and tree-based ensemble methods such as Random Forest and Gradient Boosting/XGBoost as advanced alternatives. For the operational workflow, Random Forest was selected as the primary scoring model because it can model nonlinear interactions among logistics, market, customer, and order features while also producing class probabilities and feature-importance measures.

The model output is not limited to a binary late/not-late label. Each record receives an estimated Late Delivery Probability between 0 and 1. This probability is subsequently transformed into an operational Risk Category so that the output can be consumed by non-technical stakeholders in a dashboard.

4.6 Model Evaluation Framework

The model-development requirement uses ROC-AUC, Precision, Recall, F1 Score, and the Confusion Matrix. Each metric has a different business interpretation. ROC-AUC evaluates overall discrimination; Precision indicates how many flagged high-risk cases are genuinely late; Recall measures how many actual late deliveries the model successfully identifies; F1 balances Precision and Recall; and the Confusion Matrix shows the distribution of correct and incorrect classifications.

This paper does not report unverified model-performance figures. Final performance values should be included only from the validated model run used to generate the scored dataset. This avoids overstating model quality and preserves methodological integrity.

5. DESCRIPTIVE ANALYSIS AND RESULTS

5.1 Overall Late-Delivery Incidence

The descriptive analysis identifies 98,977 late-delivery records out of 180,519 analytical records. This corresponds to an overall late-delivery rate of approximately 54.83%. Therefore, more than half of the observed order-item records are associated with a late-delivery outcome.

This high baseline rate establishes late delivery as a material operational issue rather than a small exception. It also implies that the business value of risk prediction lies not merely in identifying whether delays exist, but in ranking and prioritizing the records where intervention is most useful.

5.2 Shipping-Mode Analysis

Shipping Mode	Total Records	Late Records	Late Rate
First Class	27,814	26,513	95.32%
Second Class	35,216	26,987	76.63%
Same Day	9,737	4,454	45.74%
Standard Class	107,752	41,023	38.07%
Overall	180,519	98,977	54.83%

The strongest descriptive finding is the very high late-delivery rate associated with First Class shipments. Approximately 95.32% of First Class records are late. Second Class is also high at 76.63%. In contrast, Same Day records a 45.74% late rate and Standard Class 38.07%.

The results reveal an important operational distinction between late rate and late volume. Standard Class has the lowest late rate among the four modes, but because it handles the largest volume, it still produces 41,023 late records, the largest absolute late count. First Class, on the other hand, has a smaller total volume but an extremely high failure rate against its delivery commitment. Effective operations management therefore needs to monitor both percentage risk and absolute workload.

5.3 Business Interpretation of Shipping-Mode Risk

The shipping-mode pattern suggests that tighter promised service windows can create elevated risk if actual logistics capability is not consistently aligned with those commitments. A premium or faster service label should not automatically be interpreted as a lower risk of missing the promise. From an operational perspective, First Class and Second Class require close risk monitoring because their historical late rates are substantially above the overall baseline.

Standard Class requires a different management response. Its rate is comparatively lower, but its volume makes it a major contributor to the total pool of delayed records. This demonstrates why a dashboard should provide both risk-rate KPIs and a high-risk action queue instead of relying on a single percentage.

5.4 Region, Market and Customer-Segment Analysis

The Tableau workflow extends the descriptive analysis beyond shipping mode. Risk is examined by Order Region, Market, Customer Segment, and combinations of Region $\times$ Shipping Mode. The purpose of these views is to identify concentrations of predicted probability rather than only historical late counts. This distinction is important because the final dashboard is designed as a forward-looking operational tool.

A heatmap using Order Region on one dimension and Shipping Mode on the other uses average Late_Probability as the color and label measure. This enables operations users to quickly identify combinations of geography and service mode that require attention. The dashboard also provides market and customer-segment risk views, allowing risk patterns to be compared across different business slices.

6. RISK PROBABILITY AND OPERATIONAL CLASSIFICATION

6.1 Late Delivery Probability

The central model output is Late_Probability, a continuous score between 0 and 1 that represents the estimated probability that a record will experience late delivery. A probability score is operationally more useful than a simple binary prediction because it allows records to be ranked according to urgency.

6.2 Risk Categories

For dashboard consumption, probability scores are grouped into three operational categories: Low Risk, Medium Risk, and High Risk. The project workflow initially used the following practical thresholds: Low Risk up to 0.30, Medium Risk from above 0.30 to 0.60, and High Risk above 0.60. These thresholds can be tuned later according to business tolerance for false alarms, intervention cost, and the desired level of recall.

Risk Category	Probability Range	Operational Meaning
Low Risk	0.00–0.30	Routine monitoring; limited intervention required.
Medium Risk	>0.30–0.60	Review relevant risk drivers and operational context.
High Risk	>0.60–1.00	Prioritize for immediate operational attention.

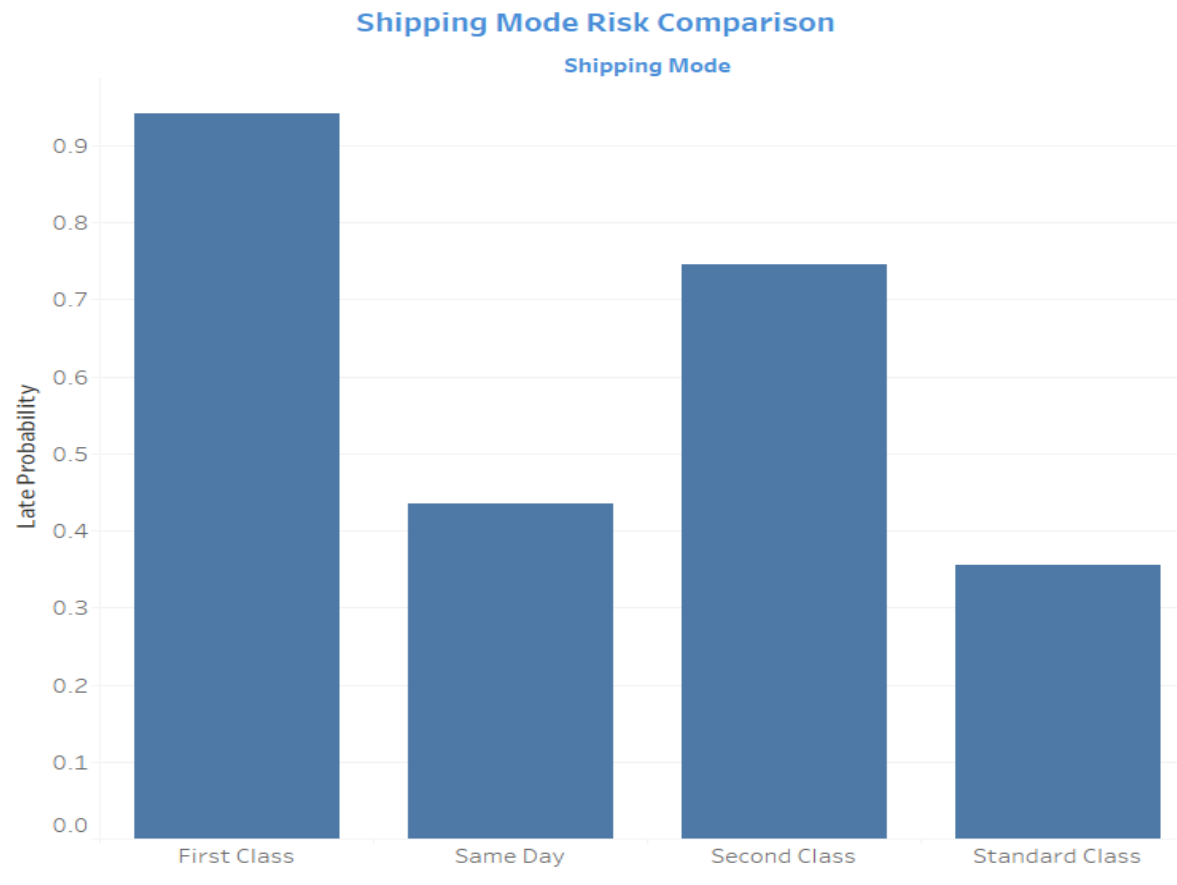


Figure :1 Shipping Mode

6.3 Regional Shipping Mode Analysis

Regional shipping mode analysis was conducted to understand how delivery risk varies across different geographical regions and transportation modes. The purpose of this analysis is to identify whether specific combinations of region and shipping mode are associated with comparatively higher late-delivery risk.

The chart compares shipping modes across different regions and highlights variations in delivery performance. The analysis helps identify areas where a particular shipping mode may experience higher operational pressure or greater probability of delay.

The findings indicate that late-delivery risk is not uniform across all regions and shipping modes. Certain region–mode combinations demonstrate relatively higher delivery risk, suggesting that both geographical conditions and the selected shipping method can influence shipment performance.

Regional Shipping Mode Risk Heatmap

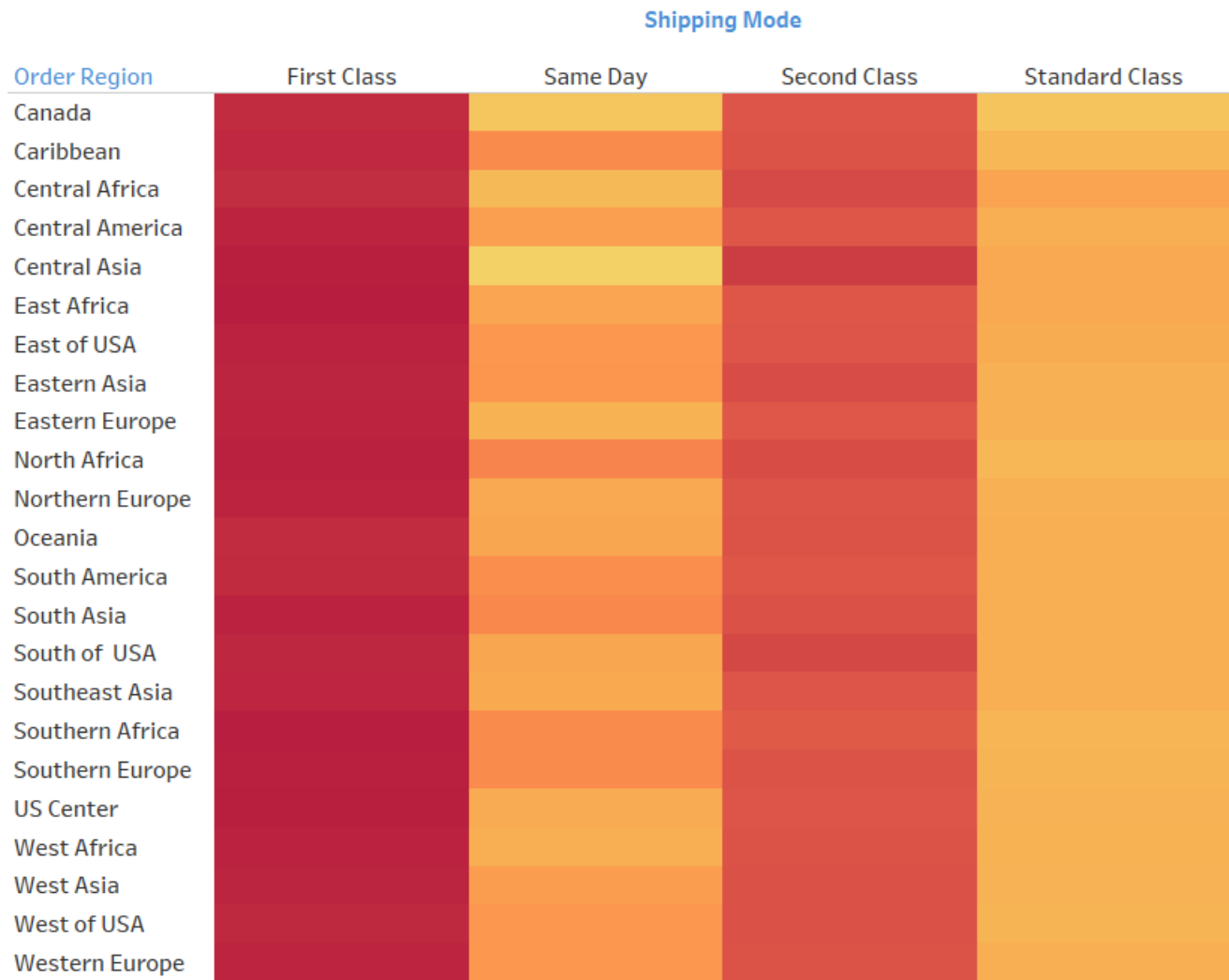


Figure: Regional Shipping Mode Risk Analysis

6.3 High-Risk Order Action Panel

The high-risk order table converts the predictive model into an operational action queue. Instead of presenting only aggregate charts, the table displays individual high-risk records and their logistics context. Fields used in the action view include row/order identifier, Order Region, Market, Shipping Mode, Customer Segment, Late Probability, and Risk Category.

The table is sorted so that the highest predicted probabilities appear first. This design allows an operations user to start with the most urgent records, investigate relevant mode and regional factors, and take action such as reprioritization, proactive communication, or further shipment review.

7. TABLEAU DASHBOARD DESIGN

7.1 Dashboard 1: Delay Risk Overview

The Delay Risk Overview provides a management-level summary of predictive risk. Core elements include Average Late Probability, High-Risk Order Count, overall risk-category distribution, and interactive filters. Its purpose is to answer: how much delivery risk currently exists, and how many records require priority attention?

7.2 Dashboard 2: Region & Mode Risk Analysis

The Region & Mode Risk Analysis focuses on the operational drivers and concentrations of risk. It compares risk across shipping modes, order regions, markets, and customer segments. A Region × Shipping Mode heatmap displays average Late_Probability and makes risky combinations visually identifiable.

7.3 Dashboard 3: Operations Action Panel

The Operations Action Panel is designed for execution rather than summary reporting. It presents the prioritized high-risk record table, enabling users to move from aggregate insight to individual action. The dashboard can be filtered by market, region, shipping mode, customer segment, and risk category so that different operational teams can focus on their relevant workload.

7.4 Dashboard Value

Together, the three dashboards create a layered decision process: the first dashboard identifies the scale of risk, the second explains where risk is concentrated, and the third identifies which records should be reviewed first. This structure aligns predictive analytics with operational workflow rather than treating machine learning as a separate technical exercise.

8. KEY FINDINGS AND INSIGHTS

1. Late delivery is a major issue in the dataset: approximately 54.83% of records are marked late.
2. First Class has the highest late-delivery rate at 95.32%, indicating a very high risk of missing the promised service level.

3. Second Class also shows elevated risk, with a 76.63% late rate.
4. Standard Class has the lowest late rate among the four modes at 38.07%, but its high volume creates the largest absolute number of late records (41,023).
5. Percentage risk and operational workload are different measures; both are required for resource allocation.
6. A probability-based model supports prioritization more effectively than a simple retrospective late/not-late report.
7. Region, market, shipping mode, and customer segment are useful operational dimensions for investigating predicted risk.
8. An order-level high-risk queue creates a direct bridge between analytics and operational intervention.
9. Leakage-aware feature selection is essential; outcome-revealing variables can make a model appear stronger than it would be in real pre-shipment use.

9. BUSINESS RECOMMENDATIONS

9.1 Prioritize First Class and Second Class Risk Review

Because First Class and Second-Class exhibit very high historical late rates, these modes should receive stronger pre-shipment risk controls. High-probability records in these modes can be routed to an exception-management queue before fulfillment proceeds.

9.2 Manage Standard Class by Volume as Well as Rate

Standard Class produces the largest number of late records because of its dominant transaction volume. Operations teams should therefore avoid interpreting its lower percentage as low business importance. Capacity planning and intervention rules should consider both late probability and shipment volume.

9.3 Use Risk Thresholds for Resource Allocation

Not every order can receive the same level of intervention. Low-, Medium-, and High-Risk categories provide a practical way to allocate attention. High-Risk records can be reviewed first, Medium-Risk records can be monitored using business rules, and Low-Risk records can follow the normal process.

9.4 Investigate Region × Shipping Mode Combinations

The heatmap should be used to identify combinations where average predicted risk is consistently high. This can support targeted investigation of carrier performance, service promises, routing constraints, or regional operational bottlenecks.

9.5 Create an Early-Warning Operating Routine

The predictive output is most valuable when embedded into a recurring operating process. The high-risk queue should be reviewed at a defined point before shipment execution, with clear ownership for decisions such as reprioritization, escalation, customer communication, or service-mode adjustment.

9.6 Monitor Model Quality and Drift

Predictive performance can change over time as customer mix, regions, products, and logistics conditions change. ROC-AUC, Precision, Recall, F1, and the confusion matrix should be monitored periodically. Risk thresholds should also be reviewed to ensure that the number of alerts remains operationally manageable.

10. LIMITATIONS AND FUTURE SCOPE

10.1 Dataset and Context Limitation

The project uses a public benchmark dataset rather than proprietary APL Logistics transaction data. Consequently, findings should be interpreted as an applied analytics demonstration in the APL Logistics business context, not as an audit of the company's actual operational performance.

10.2 Order-Item Grain

The dataset is structured at order-item level. Multiple rows may therefore relate to the same broader customer order. Future work can aggregate and validate predictions at unique order or shipment level where operational systems require one alert per shipment.

10.3 Leakage and Point-in-Time Availability

The most important technical limitation is the risk of using features that become available only after shipment or delivery. Future versions should maintain a formal point-in-time feature register documenting exactly when each predictor becomes available.

10.4 Model Validation

This paper intentionally avoids reporting model evaluation values that have not been verified from the final scoring run. The final submission can be strengthened by adding the validated confusion matrix, ROC-AUC, Precision, Recall, F1 Score, and feature-importance results from the exact model used to create Late Probability.

10.5 Explainability and Deployment

Future development can add SHAP-based local explanations so that each high-risk record includes a small set of contributing factors. A Streamlit application can also be used to make the model interactive through shipping-mode, market, customer-segment, and risk-threshold controls.

11. CONCLUSION

This research demonstrates an end-to-end framework for predicting and managing late-delivery risk in global supply-chain operations. Using the public DataCo SMART Supply Chain dataset, the project combines SQL-based validation and exploratory analysis with machine-learning risk scoring and Tableau-based operational visualization.

The descriptive results establish a significant late-delivery problem, with approximately 54.83% of records marked late. Shipping mode shows a particularly strong pattern: First Class records a 95.32% late rate and Second Class 76.63%, whereas Same Day and Standard Class record 45.74% and

38.07%, respectively. At the same time, Standard Class generates the highest absolute number of late records because of its much larger volume.

The predictive component extends this analysis from historical explanation to forward-looking risk intelligence. Each record receives a Late Delivery Probability and is translated into an operational risk category. The Tableau solution then organizes these predictions into a Delay Risk Overview, a Region & Mode Risk Analysis dashboard, and an Operations Action Panel containing the high-risk queue.

The core managerial implication is that supply-chain analytics should not stop at identifying past delays. A well-designed predictive workflow can help teams determine where risk is concentrated, which records require attention first, and how limited operational resources can be deployed more effectively. In this way, the project transforms late-delivery management from a reactive reporting problem into a proactive decision-support process.